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unun-vna 0.5.1

unun-vna is a command-line tool for measuring the **small-signal efficiency** of HF UnUn transformers with an S-A-A-2 / NanoVNA V2 using the **one-transformer method**.

The tool is generalized:

- no nominal load/series resistor is assumed;
- the actual series-load fixture is measured as a complex impedance versus frequency;
- transformer efficiency is calculated from measured S11/S21 and the measured fixture;
- transformer impedance ratio is configurable;
- 1:49 remains the default only for convenience.

The lower-impedance side of the transformer is connected to Port 1. The higher-impedance side is connected to the series fixture and then Port 2.

1. Transformer ratio

--ratio means the **impedance** ratio

$$N = \frac{Z_{\text{high}}}{Z_{\text{low}}}.$$

These are equivalent:

--ratio 49
--ratio 1:49
--ratio 1/49

Examples:

The default is $N=49$.

2. Generic measurement topology

Z_0 is normally $50\ \Omega$.

It is measured as

3. Ideal matching load for a selected ratio

$$R_{\text{high,target}} = NZ_0.$$

Port 2 itself contributes Z_0 , so the ideal purely resistive series fixture would be

At $Z_0 = 50 \ \Omega$:

These numbers are **diagnostics only**. They are never substituted for the measured fixture impedance.

4. Fixture characterization

For a pure series impedance between two equal Z_0 ports:

$$S_{11} = \frac{Z_s}{2Z_0 + Z_s}$$

$$S_{21} = \frac{2Z_0}{2Z_0 + Z_s}.$$

Therefore:

$$Z_{s,1} = 2Z_0 \frac{S_{11}}{S_{21}}$$

and

$$Z_{s,2} = 2Z_0 \left(\frac{1}{S_{21}} - 1 \right).$$

A pure series element also satisfies

$$S_{11} + S_{21} = 1.$$

The fixture command reports both impedance estimates, their disagreement, and $|S_{11}+S_{21}-1|$. This checks whether the physical fixture can safely be represented by one complex series impedance.

5. Efficiency does not depend on the configured ratio

By S-parameter definition,

$$|S_{21}|^2$$

is the incident Port-1 power fraction delivered to the matched Port-2 termination Z_0 .

The same current flows through the Port-2 resistance and the series fixture. Thus total real high-side load power is

$$\frac{P_{\text{high}}}{P_{\text{incident}}} = |S_{21}|^2 \left(1 + \frac{\Re\{Z_s\}}{Z_0} \right).$$

Power accepted at Port 1 is

$$\frac{P_{\text{accepted}}}{P_{\text{incident}}} = 1 - |S_{11}|^2.$$

Therefore

$$\eta = \frac{|S_{21}|^2 \left(1 + \frac{\Re\{Z_s\}}{Z_0}\right)}{1 - |S_{11}|^2}$$

and

$$L_{\text{dB}} = -10 \log_{10}(\eta).$$

VSWR

The analysis also reports Port-1 VSWR derived directly from calibrated S11. With

$$\rho = |S_{11}|,$$

$$\text{VSWR} = \frac{1 + \rho}{1 - \rho}$$

for $\rho < 1$.

A perfect match gives VSWR = 1.0. Total reflection gives infinite VSWR. Values with $|S_{11}| \geq 1$, which can arise from residual calibration or measurement error, are reported as infinite rather than as a non-physical negative VSWR.

The tool also reports the **ideal lossless VSWR** for the configured transformer ratio with the actually measured fixture. This helps separate load mismatch from transformer loss.

There is no N in the measured efficiency equation. That is intentional.

The measured power flow already contains the transformer's actual behaviour; the ratio is not needed to reconstruct efficiency.

6. What --ratio is used for

The configured ratio is used to calculate a **lossless ideal reference** for the actual measured fixture.

The real measured high-side load is

$$Z_{\text{load}} = Z_s + Z_0.$$

An ideal lossless transformer with impedance ratio N reflects that to Port 1 as

$$Z_{\text{in}} = \frac{Z_{\text{load}}}{N}.$$

The input reflection expected from load mismatch alone is

$$\Gamma_{\text{ideal}} = \frac{Z_{\text{in}} - Z_0}{Z_{\text{in}} + Z_0}.$$

Therefore

$$|S_{11,\text{ideal}}| = |\Gamma_{\text{ideal}}|.$$

The accepted incident power of that ideal network is

$$1 - |\Gamma_{\text{ideal}}|^2.$$

Since an ideal transformer is lossless, the fraction of that real power dissipated in Port 2 is

$$\frac{Z_0}{Z_0 + \Re\{Z_s\}}.$$

Hence

$$|S_{21,\text{ideal}}|^2 = (1 - |\Gamma_{\text{ideal}}|^2) \frac{Z_0}{Z_0 + \Re\{Z_s\}}.$$

The analysis output shows measured S11/S21 beside these ideal values.

This is useful when the fixture is not exactly the perfect matching resistance for the selected ratio.

7. S-A-A-2 Port-2 load-match limitation

The S-A-A-2 is a T/R VNA, not a full bidirectional two-port VNA.

It measures S11 and forward S21, but cannot perform a complete independent Port-2 load-match correction.

High-ratio transformer/fixture combinations can be strongly reflecting when viewed from Port 2 even when Port 1 is well matched.

A THRU verification can therefore look excellent while a highly reflective DUT still has biased S21.

Recommended Port-2 pad

Use a fixed 50 Ω attenuator, typically 10 dB, immediately before Port 2:

Port 2 ---- 10 dB pad ---- calibration plane ---- DUT

The same pad must remain in place during:

1. calibration;
2. fixture characterization;
3. transformer measurement.

The THRU calibration removes its forward loss and phase. Reflections from Port 2 pass through it twice, so a 10 dB pad reduces the load-match influence by roughly 20 dB.

8. Calibration

```
unun-vna calibrate \
--start 1M \
--stop 35M \
--points 501 \
-o hf35-pad
```

The program requests:

1. SHORT on Port 1;
2. OPEN on Port 1;
3. 50 Ω LOAD on Port 1;
4. THRU between the calibration planes.

Files:

hf35-pad.npz
hf35-pad.npz.json
hf35-pad.npz.through.csv

9. Characterize a fixture

The fixture extraction itself is independent of ratio.

For a 1:9 reference:

```
unun-vna fixture \  
  --cal hf35-pad.npz \  
  --ratio 1:9 \  
  -o fixture-1to9
```

For 1:64:

```
unun-vna fixture \  
  --cal hf35-pad.npz \  
  --ratio 64 \  
  -o fixture-1to64
```

--ratio changes only the target-load diagnostics printed by this command.

10. Measure a transformer

Measurement acquisition itself requires no ratio:

```
unun-vna measure \  
  --cal hf35-pad.npz \  
  -o transformer
```

The measured chain is:

Port 1 ---- transformer ---- characterized fixture ---- Port 2 pad ---- Port 2

11. Analyze

Default 1:49:

```
unun-vna analyze \  
  --dut transformer.npz \  
  --fixture fixture.npz \  
  -o efficiency.csv
```

Explicit 1:49:

```

unun-vna analyze \
  --dut transformer.npz \
  --fixture fixture.npz \
  --ratio 1:49 \
  -o efficiency.csv

1:9:
unun-vna analyze \
  --dut transformer.npz \
  --fixture fixture-1to9.npz \
  --ratio 1:9 \
  -o efficiency-1to9.csv

1:64:
unun-vna analyze \
  --dut transformer.npz \
  --fixture fixture-1to64.npz \
  --ratio 64 \
  -o efficiency-1to64.csv

```

12. Analysis CSV

The analysis CSV contains:

```

frequency_hz
s11_db
vswr
s21_db
fixture_r_ohm
fixture_x_ohm
impedance_ratio_high_over_low
target_high_side_resistance_ohm
target_series_resistance_ohm
ideal_lossless_s11_db_for_measured_fixture
ideal_lossless_vswr_for_measured_fixture
ideal_lossless_s21_db_for_measured_fixture
accepted_fraction
delivered_fraction
efficiency
efficiency_percent
transformer_loss_db

```

The target_* and ideal_lossless_* columns depend on --ratio.

The actual efficiency columns do not.

13. Ideal sanity checks

1:9

At $Z_0=50\ \Omega$, use an ideal $400\ \Omega$ series fixture:

$$400 + 50 = 450\ \Omega$$

and

$$450/9 = 50 \, \Omega.$$

Thus an ideal transformer has

$$S_{11} = 0$$

and

$$|S_{21}| = 1/3$$

or approximately

$$S_{21} = -9.54 \, \text{dB}.$$

Efficiency evaluates to 100%.

1:49

With 2400 Ω series:

$$|S_{21}| = 1/7$$

or

$$S_{21} \approx -16.90 \, \text{dB}.$$

1:64

With 3150 Ω series:

$$|S_{21}| = 1/8$$

or

$$S_{21} \approx -18.06 \, \text{dB}.$$

14. Remaining assumptions

The tool does **not** assume:

- a 1:49 transformer;
- a 2400 Ω fixture;
- any nominal fixture resistance;
- that the fixture is purely resistive.

It does assume:

1. the lower-impedance transformer side is connected to Port 1;
2. the fixture is adequately represented by one complex series impedance;
3. Port 2 is a matched real Z_0 termination at the calibrated reference plane;

4. calibration, fixture measurement and DUT measurement use the same RF setup;
5. a fixed Port-2 pad, if used, remains in place;
6. the measurement is small-signal.

A fully arbitrary two-port fixture would require S12 and S22 and therefore a full bidirectional VNA.

15. Full workflow example

```
pipx install --editable . --force
```

```
unun-vna info
```

```
unun-vna calibrate \  
  --start 1M \  
  --stop 35M \  
  --points 501 \  
  -o hf35-pad
```

```
unun-vna fixture \  
  --cal hf35-pad.npz \  
  --ratio 1:49 \  
  -o fixture
```

```
unun-vna measure \  
  --cal hf35-pad.npz \  
  -o transformer
```

```
unun-vna analyze \  
  --dut transformer.npz \  
  --fixture fixture.npz \  
  --ratio 1:49 \  
  -o efficiency.csv
```

16. Source structure

```
unun-vna/  
├── pyproject.toml  
├── README.md  
├── src/  
│   └── unun_vna/  
│       ├── __init__.py  
│       ├── cli.py  
│       └── analysis.py  
└── tests/  
    └── test_math.py
```

`analysis.py` deliberately separates:

- ratio-independent measured efficiency;
- ratio-dependent ideal-reference diagnostics.

This prevents a nominal transformer ratio from silently changing the measured power calculation.